

SMART POLYPHARMACY AND VITAMIN RECOMMENDATION SYSTEM

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BSc (Hons) degree in Information Technology Specializing in
Information Technology

Department of Computer Science

Sri Lanka Institute of Information Technology Sri Lanka

APRIL 2026

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DECLARATION

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ABSTRACT

The increasing prevalence of polypharmacy and vitamin deficiencies, particularly among geriatric populations, has become a significant concern in modern healthcare. The simultaneous use of multiple medications can lead to adverse drug interactions, reduced treatment effectiveness, and increased health risks. In parallel, undiagnosed or late-detected vitamin deficiencies can contribute to chronic health conditions, reduced immunity, and decreased quality of life. Addressing these challenges requires an integrated, intelligent, and personalized approach.

This research presents the development of a *Smart Polypharmacy and Vitamin Recommendation System*, designed to assist in early detection, risk assessment, and personalized health guidance. The system consists of four main components. The first component is an early vitamin deficiency warning system that analyzes user-reported symptoms alongside long-term medication usage to identify potential deficiencies at an early stage. The second component is a polypharmacy risk score system that evaluates potential drug interactions and calculates a risk score to highlight possible adverse effects.

The third component focuses on providing personalized mental health and lifestyle advisory support for geriatric patients, considering factors such as medication burden, daily habits, and emotional well-being. The fourth component is a personalized food recommendation system that suggests dietary plans tailored to address identified vitamin deficiencies, promoting natural and sustainable health improvements.

The proposed system leverages data-driven techniques and intelligent decision-making models to deliver accurate, user-centric recommendations. By integrating multiple healthcare aspects into a single platform, the system aims to improve patient safety, enhance early diagnosis, and support overall well-being. This research contributes to the growing field of smart healthcare solutions by providing a comprehensive and scalable approach to managing polypharmacy and nutritional health.

Keywords: Polypharmacy, Vitamin Deficiency, Drug Interaction, Personalized Healthcare, Geriatric Care, Recommendation System, Smart Health Systems

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
SVM	Support Vector Machine
LR	Logistic Regression
OvR	One-Vs-Rest
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
API	Application Programming Interface
F1- score	Harmonic Mean of Precision and Recall
KNN	K-Nearest Neighbors
NB	Naïve Bayes
DL	Deep Learning
BERT	Bidirectional Encoder Representations from Transformers

1. INTRODUCTION

The global population is rapidly aging, with the number of individuals aged 60 years and above expected to increase significantly in the coming decades. This demographic shift has led to a rise in chronic diseases such as diabetes, cardiovascular disorders, and neurological conditions, particularly among elderly populations. As a result, many older adults require multiple medications simultaneously, a condition known as polypharmacy. Polypharmacy, commonly defined as the concurrent use of five or more medications, has become a widespread phenomenon in geriatric healthcare and is associated with increased risks of adverse drug reactions, hospitalizations, and reduced quality of life [1], [2].

One of the most critical yet often overlooked consequences of polypharmacy is drug–drug and drug–nutrient interactions. These interactions can significantly alter the effectiveness of medications and impair the absorption and metabolism of essential nutrients. Long-term use of commonly prescribed drugs has been shown to contribute to deficiencies in vital vitamins such as B12, D, and folate, leading to symptoms including fatigue, anemia, cognitive decline, osteoporosis, and weakened immunity [1], [2]. Despite these risks, traditional healthcare approaches often focus on treating diseases rather than preventing such deficiencies at an early stage.

Early detection of vitamin deficiencies remains a major challenge in clinical practice. Conventional diagnostic methods typically rely on laboratory testing after symptoms become severe, resulting in delayed intervention. Recent advancements in artificial intelligence (AI) and machine learning (ML) have introduced new possibilities for proactive healthcare. AI-based systems, including rule-based models, expert systems, and neural networks, have demonstrated the ability to analyze symptoms and predict potential deficiencies before they escalate into serious conditions [3], [4]. However, many of these systems are limited to detection and do not provide actionable solutions such as dietary recommendations tailored to individual patient needs.

At the same time, existing diet and nutrition applications primarily focus on calorie tracking and general health management. While useful for the broader population, these systems lack clinical depth and fail to consider critical factors such as medication history, drug–nutrient interactions, comorbidities, and patient-specific characteristics like Body Mass Index (BMI) and lifestyle patterns. Consequently, they are not well-suited for elderly patients who require personalized and medically informed nutritional guidance.

In addition to physical health challenges, geriatric populations face significant mental health concerns. Studies indicate that a considerable proportion of older adults experience anxiety, depression, sleep disorders, and cognitive decline, which are often exacerbated by chronic illness and medication burden [4]. Despite this, most digital health solutions focus narrowly on medication adherence and symptom monitoring, overlooking the importance of mental well-being and lifestyle management. This gap highlights the need for integrated systems that address both physical and psychological aspects of elderly care.

To address these challenges, this research proposes a comprehensive *Smart Polypharmacy and Vitamin Recommendation System* that integrates multiple healthcare components into a unified platform. The system is designed with four key modules. The first module is an Early Vitamin Deficiency Warning System that utilizes symptoms and

long-term medication data to identify potential deficiencies at an early stage. The second module is a Polypharmacy Risk Score System that evaluates drug–drug interactions and predicts potential risks using data-driven techniques. The third module focuses on providing personalized mental health and lifestyle recommendations tailored to geriatric patients. The fourth module is a Personalized Food Recommendation System that generates diet plans aimed at correcting vitamin deficiencies while considering individual preferences, medical conditions, and ongoing treatments.

Unlike existing solutions, the proposed system adopts a holistic and personalized approach by combining predictive analytics, healthcare informatics, and intelligent recommendation techniques. It not only identifies risks but also provides actionable interventions, including dietary and lifestyle guidance, while ensuring compatibility with ongoing medical treatments. By integrating these components, the system aims to improve early diagnosis, enhance patient safety, and support overall well-being among elderly individuals.

In conclusion, while previous research has explored various approaches to vitamin deficiency detection, polypharmacy risk assessment, and dietary recommendation, these efforts are often fragmented and limited in scope. This study bridges these gaps by developing an integrated, adaptive, and patient-centered system that addresses the complex healthcare needs of geriatric populations. The proposed solution represents a significant step toward smarter, safer, and more personalized healthcare management of severe complications and contribute to more proactive healthcare management.

1.1. Background Literature

The growing complexity of modern healthcare systems has led to increased research interest in intelligent decision-support systems for managing polypharmacy and nutritional deficiencies. Traditionally, clinical decision-making for vitamin deficiencies relied on laboratory tests and physician assessments. Early computational approaches included rule-based expert systems that attempted to replicate clinical reasoning by mapping symptoms to predefined diagnostic rules. While such systems provided structured reasoning capabilities, they were limited by their rigidity and inability to generalize beyond predefined knowledge bases [3], [4]. These limitations highlighted the need for more adaptive and data-driven approaches in healthcare diagnostics.

With the emergence of artificial intelligence (AI) and machine learning (ML), researchers began exploring automated methods for disease and deficiency detection. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated strong performance in classifying medical conditions from complex datasets, including symptom-based and image-based inputs [1], [2]. These models are capable of learning hidden patterns in data without explicit feature engineering, making them highly effective for medical prediction tasks. However, their dependency on large labeled datasets and high computational requirements restrict their applicability in many real-world clinical environments.

In parallel, classical machine learning techniques have been widely applied to healthcare prediction problems due to their interpretability and lower data requirements. Studies using multi-classifier frameworks combining algorithms such as Support Vector Machines (SVM), Decision Trees, and Naïve Bayes have shown promising results in

vitamin deficiency detection and nutritional recommendation tasks [5]. These models are particularly useful in structured medical datasets where feature relationships are relatively well-defined, offering a balance between performance and interpretability.

From a biomedical perspective, vitamin deficiencies remain a significant global health concern. Vitamin B12 deficiency is strongly associated with neurological impairment and anemia, while Vitamin A deficiency affects immune function and vision, and Vitamin D deficiency is linked to bone disorders and chronic diseases [6], [7], [8]. Research also indicates that vitamins play a broader role in preventing complex conditions such as neurodegenerative diseases and inflammatory disorders, emphasizing the importance of early detection and preventive care [9], [10]. These findings reinforce the need for intelligent systems capable of continuous monitoring and early warning of deficiency risks, especially in vulnerable populations such as the elderly.

Polypharmacy has emerged as a critical contributing factor to vitamin deficiencies in geriatric patients. The simultaneous use of multiple medications increases the risk of drug–drug and drug–nutrient interactions, which can impair nutrient absorption and metabolic processes. Studies have shown that elderly patients frequently experience adverse outcomes due to inappropriate medication combinations and cumulative drug effects [2], [12]. Additionally, the global rise in polypharmacy prevalence highlights the urgency of developing predictive systems that can assess medication-related risks and support safer prescribing practices [13].

Recent advancements in healthcare analytics have introduced predictive models for polypharmacy risk assessment. Machine learning approaches have been used to identify high-risk patients based on medication patterns, comorbidities, and clinical history [15], [18]. Some studies have also explored pharmacokinetic–pharmacodynamic modeling and drug–drug interaction prediction using computational techniques [11]. Despite these advancements, many existing systems remain limited in their ability to integrate nutritional consequences such as vitamin depletion, leaving a critical gap in holistic patient management.

Alongside physical health concerns, geriatric populations also face significant mental health and lifestyle challenges. Research indicates high prevalence of anxiety, depression, sleep disorders, and cognitive decline among older adults, which are often worsened by chronic illness and medication burden [4], [5]. Digital health interventions such as AI-based companions, socially assistive robots, and mindfulness systems have shown potential in improving emotional well-being and sleep quality among elderly individuals [6], [9], [10]. However, most of these systems operate independently of medical and nutritional monitoring, lacking integration with clinical decision-making frameworks.

In the domain of nutritional healthcare, AI-driven food recommendation systems have gained increasing attention. Studies have demonstrated the use of machine learning and optimization techniques such as genetic algorithms for generating personalized meal plans [7], [8]. More recent research has focused on culturally adaptive and user-specific dietary recommendations, particularly for elderly populations [9], [10], [11]. These systems aim to improve dietary behavior and health outcomes, but many are still generalized and do not specifically address vitamin deficiencies caused by long-term medication use.

Existing literature also highlights the importance of integrating drug–nutrient interactions into dietary recommendation systems. Research shows that chronic medication usage can significantly alter nutrient absorption, leading to long-term deficiencies if not properly managed [1]. Despite this, most current nutritional systems fail to incorporate medication history or interaction effects into their recommendation logic, limiting their effectiveness in clinical scenarios.

To address these gaps, the proposed *Smart Polypharmacy and Vitamin Recommendation System* integrates four interconnected components: (1) an Early Vitamin Deficiency Warning System based on symptoms and long-term medication usage, (2) a Polypharmacy Risk Score System based on drug–drug interaction analysis, (3) a Personalized Mental Health and Lifestyle Advisory System for geriatric patients, and (4) a Personalized Food Recommendation System for vitamin deficiency management. This integrated approach aligns with recent trends in AI-driven personalized healthcare, aiming to move from isolated diagnostic tools toward comprehensive, patient-centric decision-support systems.

By combining insights from AI-based diagnosis, pharmacological interaction modeling, mental health support systems, and nutritional recommendation research, this study positions itself within the emerging field of holistic digital healthcare. It addresses the limitations of existing approaches by focusing on integration, personalization, and real-world applicability in geriatric care environments.

1.2. Research Gap

3. RESEARCH GAP

Despite significant advancements in artificial intelligence, machine learning, and healthcare informatics, existing studies still present several limitations when applied to real-world geriatric healthcare scenarios. The comparative analysis of prior research highlights clear gaps across vitamin deficiency detection, polypharmacy risk assessment, mental health integration, and personalized food recommendation systems.

3.1 Gap in Vitamin Deficiency Detection Systems

Existing vitamin deficiency detection approaches are mainly based on image processing techniques such as Convolutional Neural Networks (CNNs) and neural networks applied to visual symptoms [1], [2]. While these models achieve high accuracy in controlled environments, they rely heavily on image-based inputs such as skin, nails, or facial features. Traditional expert systems using forward chaining also attempt symptom-based diagnosis [3], [4], but they depend on static rule sets.

However, all these approaches lack integration with long-term medication history and drug-induced nutrient depletion patterns. As shown in Table 1, none of the existing systems fully support multi-vitamin detection combined with medication-based analysis. This creates a critical gap in early warning systems for patients undergoing long-term pharmacological treatment.

3.2 Gap in Polypharmacy Risk Assessment Systems

Existing polypharmacy risk models primarily rely on simple drug count-based scoring or limited patient-specific attributes such as age or comorbidities [15], [18]. Some studies incorporate clinical decision support systems for alerts, but most do not consider Drug–Drug Interaction (DDI) severity in a comprehensive manner.

Furthermore, many models are static and do not adapt dynamically to changes in prescriptions or patient health conditions. As shown in comparative studies (Research A–D), there is limited integration of ML-driven predictive scoring with real-time clinical decision support. This results in a gap for developing adaptive, AI-powered polypharmacy risk scoring systems that can provide continuous and personalized risk evaluation.

3.3 Gap in Mental Health and Lifestyle Integration

Although recent research highlights the importance of mental health support for elderly patients, existing digital health systems largely focus on medication adherence and physical health monitoring [4], [5]. AI-based solutions such as social robots and emotional intelligence systems have been proposed [9], [10], but they remain isolated from clinical decision-making frameworks.

Current systems fail to integrate emotional well-being, lifestyle behavior, and medication burden into a unified advisory model. This creates a gap in holistic elderly care systems that simultaneously address mental health, lifestyle guidance, and pharmacological safety.

3.4 Gap in Personalized Food Recommendation Systems

Existing food recommendation systems primarily focus on general nutrition planning, calorie tracking, or disease-specific diets [7], [8]. However, they do not adequately incorporate drug–nutrient interactions or long-term medication effects on vitamin absorption (Research A). Additionally, most systems are designed for general populations and do not specifically address geriatric polypharmacy cases (Research B).

Another major limitation is the lack of dynamic adaptation in dietary recommendations. Current models generate static meal plans and fail to update based on changes in medication or health status (Research C). Furthermore, cultural preferences, socioeconomic factors, and individual dietary habits are often ignored, reducing user compliance and effectiveness (Research D).

3.5 Summary of Research Gap

Based on the comparative analysis, it is evident that existing systems are fragmented and limited in scope. They either focus on single aspects such as vitamin detection, drug interaction analysis, or food recommendation, without integrating them into a unified framework. There is a clear need for a comprehensive, adaptive, and intelligent system that combines:

- Symptom and medication-based vitamin deficiency detection
- AI-driven polypharmacy risk scoring with DDI severity
- Mental health and lifestyle advisory for geriatric patients
- Personalized, culturally aware food recommendation systems

The proposed *Smart Polypharmacy and Vitamin Recommendation System* addresses these gaps by integrating all four components into a unified, data-driven healthcare solution aimed at improving elderly patient safety and overall well-being.

Feature / Aspect	Research [1](CNN Image Detection)	Research [2](Neural Network Image Based)	Research [3](Expert System Forward Chaining)	Research [4](Expert System Mobile Based)	Proposed Solution (Symptom + Medication-Based Warning)
Multi-vitamin detection	No	No	No	No	Yes
Integration	No	No	No	No	Yes

with long term medication					
Symptom-based early warning	No	No	Yes	Yes	Yes
Predictive risk scoring	No	No	No	No	Yes
Personalized preventive Guidance	No	No	No	Yes	Yes
AI / ML-driven analysis	Yes	Yes	No	Partially	Yes
Mobile / web accessibility	No	No	No	Yes	Yes
Adaptive system learning	No	No	No	No	Yes
Non-image input (symptoms & history)	No	No	No	No	Yes
Multi-factor health data (diet, lifestyle, medication)	No	No	No	No	Yes

Table 1: Comparison between existing systems for vitamin deficiency prediction

Table 2: Comparison between existing systems for risk
Table 3: Comparison between existing systems for food
Table 4: Comparison between existing systems for mental health

1.3. Research Problem

Despite the rapid advancement of artificial intelligence and digital health technologies, current healthcare systems remain fragmented when addressing the complex needs of geriatric patients. Elderly individuals who are exposed to polypharmacy are at a significantly higher risk of drug–drug interactions, adverse drug reactions, and long-term nutrient deficiencies due to impaired vitamin absorption. At the same time, existing systems for vitamin deficiency detection are largely limited to image-based or rule-based approaches, which do not consider long-term medication usage or multi-factor health data.

In addition, current polypharmacy management tools primarily focus on medication counts and basic clinical attributes, without adequately incorporating drug–drug interaction severity or providing dynamic, personalized risk scoring. Mental health and lifestyle factors, which are critical in elderly care, are also often overlooked in existing digital health solutions. Furthermore, most food recommendation systems are designed for general populations and fail to account for drug-induced nutritional deficiencies, cultural preferences, and patient-specific conditions such as BMI and comorbidities.

As a result, there is no unified system that integrates vitamin deficiency prediction, polypharmacy risk assessment, mental health advisory, and personalized nutritional guidance into a single intelligent framework. This gap limits the effectiveness of preventive healthcare and reduces the ability to provide early, personalized, and holistic interventions for elderly patients.

Therefore, this research aims to develop a *Smart Polypharmacy and Vitamin Recommendation System* that addresses these limitations by combining symptom-based analysis, medication history, drug interaction modeling, and personalized recommendation techniques to improve early detection, patient safety, and overall healthcare outcomes.

1.4. Research Objectives

The main objective of this research is to design and develop a *Smart Polypharmacy and Vitamin Recommendation System* that provides an integrated, intelligent, and personalized healthcare solution for elderly patients. The system aims to improve early detection of vitamin deficiencies, assess polypharmacy risks, and deliver personalized lifestyle and nutritional recommendations.

To achieve the main objective, the following specific objectives are identified:

- To develop an Early Vitamin Deficiency Warning System that identifies potential vitamin deficiencies based on symptoms and long-term medication usage.
- To design a Polypharmacy Risk Score System that evaluates drug–drug interactions and generates a risk score using patient medication data.
- To implement a Personalized Mental Health and Lifestyle Advisory System tailored for geriatric patients to support emotional well-being and daily lifestyle management.
- To develop a Personalized Food Recommendation System that suggests suitable dietary plans for correcting vitamin deficiencies while considering drug–nutrient

interactions and patient preferences.

- To integrate all four components into a unified system that provides comprehensive, data-driven healthcare support for elderly individuals.
- To evaluate the effectiveness of the proposed system in improving early detection, risk assessment accuracy, and personalization of recommendations.

1.4.1. Main Objectives

The main objective of this research is to design and develop a *Smart Polypharmacy and Vitamin Recommendation System* that integrates four core components: an Early Vitamin Deficiency Warning System based on symptoms and long-term medication usage, a Polypharmacy Risk Score System based on drug–drug interaction analysis, a Personalized Mental Health and Lifestyle Advisory System for geriatric patients, and a Personalized Food Recommendation System for vitamin deficiency management. The system aims to provide a unified, intelligent, and personalized healthcare solution that enhances early detection of vitamin deficiencies, improves medication safety, and supports overall physical, mental, and nutritional well-being of elderly patients.

To achieve this main objective, the study further focuses on implementing and evaluating each component to ensure accurate prediction, effective risk assessment, and meaningful personalized recommendations within a single integrated framework.

1.4.2. Sub-Objectives

- To identify and preprocess relevant datasets including symptoms, medication records, drug interactions, and nutritional information for model development.
- To perform exploration data analysis (EDA) to understand patterns in vitamin deficiencies and polypharmacy-related risks.
- To apply appropriate feature extraction techniques such as symptom encoding, medication profiling, and text vectorization for improving model input quality.
- To develop a machine learning–based approach for early vitamin deficiency prediction using symptom and medication-based data.
- To compare multiple classification algorithms (e.g., SVM, Logistic Regression, Naïve Bayes, KNN) and select the most effective model based on evaluation metrics.
- To design and implement a Polypharmacy Risk Score System by integrating drug–drug interaction severity and patient medication history.
- To incorporate drug–nutrient interaction knowledge into the system to ensure safe and effective dietary recommendations.
- Developing a Personalized Food Recommendation System that suggests suitable diets based on identified vitamin deficiencies and patient preferences.
- To design a Personalized Mental Health and Lifestyle Advisory module for geriatric patients to support emotional well-being and daily lifestyle improvements.
- To develop a risk interpretation and explanation mechanism to provide clear and understandable outputs for healthcare decision-making.
- To ensure system adaptability by enabling dynamic updates based on changes in medication, symptoms, and patient health conditions.
- To design and implement a user-friendly interface for both patients and healthcare providers to interact with the system effectively.
- To evaluate the performance of the proposed system using metrics such as accuracy, precision, recall, and F1-score.
- To ensure data privacy, security, and ethical handling of sensitive patient health information throughout the system.

2. Methodology

2.1. Methodology

The proposed *Smart Polypharmacy and Vitamin Recommendation System* is an intelligent, integrated healthcare solution designed to support elderly patients who are affected by polypharmacy and vitamin deficiencies. The system combines multiple AI-driven modules to provide early detection, risk assessment, mental health support, and personalized nutritional guidance within a single platform.

The system is structured around four main components. The first component is the *Early Vitamin Deficiency Warning System*, which analyzes patient symptoms along with long-term medication history to identify potential vitamin deficiencies at an early stage. This module helps in proactive healthcare by generating alerts before conditions become severe.

The second component is the *Polypharmacy Risk Score System*, which evaluates a patient's medication profile by analyzing drug–drug interactions and assigns a risk score. This enables healthcare providers to identify high-risk patients and take preventive actions to reduce adverse drug reactions and complications.

The third component is the *Personalized Mental Health and Lifestyle Advisory System*, which focuses on improving the emotional well-being and daily lifestyle of geriatric patients. It provides tailored suggestions based on patient conditions, medication burden, and general health behavior patterns.

The fourth component is the *Personalized Food Recommendation System*, which generates customized dietary plans based on detected vitamin deficiencies. This module also considers drug–nutrient interactions, patient preferences, and health conditions to ensure safe and effective nutritional guidance.

These four components are integrated into a unified system that processes patient data such as symptoms, medication history, and lifestyle inputs. The system applies machine learning techniques, data analysis, and recommendation algorithms to generate intelligent outputs. The final results are presented through a user-friendly interface for both patients and healthcare providers.

Overall, the system aims to shift healthcare from a reactive approach to a proactive and preventive model by enabling early detection, personalized care, and improved decision-making in geriatric health management.

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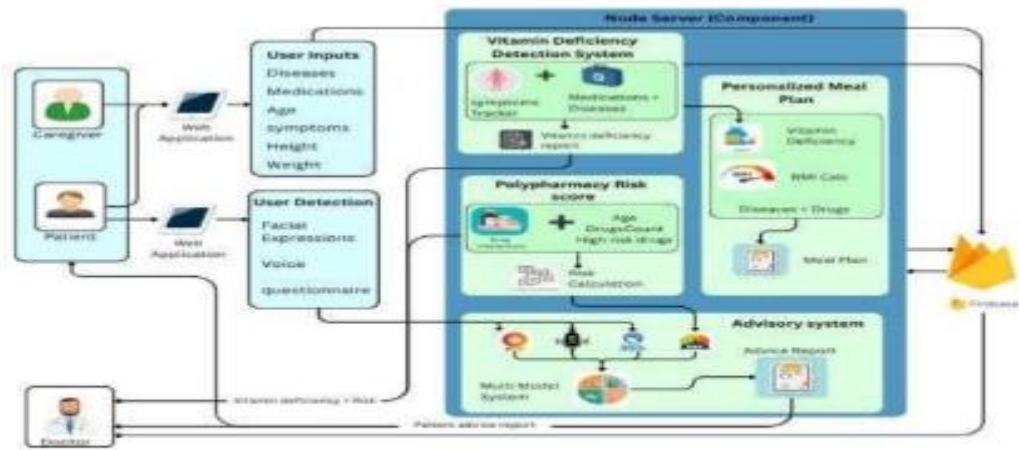


Figure 2: the system overview diagram

Figure 1: System Overview Diagram

2.1.1. Data Collection and Preprocessing

Data collection is a fundamental step in developing the *Smart Polypharmacy and Vitamin Recommendation System*, as the performance of the proposed models heavily depends on the quality and relevance of the input data. In this study, multiple data sources are considered to support the four core components of the system, including vitamin deficiency prediction, polypharmacy risk assessment, mental health advisory, and personalized food recommendation.

The primary data includes symptom-related information, long-term medication records, and drug interaction data. In addition, nutritional datasets containing vitamin-rich food information and vitamin deficiency profiles are utilized to support the recommendation module. Drug–drug interaction data is also incorporated to assess polypharmacy risks accurately. These datasets are collected from publicly available medical sources, research studies, and validated healthcare databases to ensure reliability and accuracy.

Once the data is collected, preprocessing is performed to improve data quality and make it suitable for machine learning models. This includes handling missing values, removing duplicates, and correcting inconsistent or noisy entries. Since medical data often contains unstructured or semi-structured text, text normalization techniques such as lowercasing, tokenization, and stop-word removal are applied where necessary.

For symptom and medication-based inputs, feature encoding techniques are used to convert categorical data into numerical representations. Techniques such as label encoding and vectorization (e.g., TF-IDF for text-based symptoms) are applied to transform raw data into a machine-readable format. Additionally, drug interaction severity levels are standardized to ensure consistency across the dataset.

Data integration is also performed to combine multiple sources into a unified dataset, ensuring that symptoms, medication, nutritional, and interaction data are properly aligned for each patient record. Finally, the preprocessed dataset is split into training and testing sets to support model development and evaluation.

Overall, data collection and preprocessing ensure that the system is built on clean, structured, and meaningful data, which is essential for achieving accurate predictions and reliable recommendations across all components of the proposed system.

Scenario coverage:

To ensure robustness across all four components of the *Smart Polypharmacy and Vitamin Recommendation System*, the dataset generation and collection framework is designed to represent a wide range of realistic and clinically relevant scenarios. This is essential to improve model generalization, reduce bias, and ensure reliable performance in real-world healthcare environments.

Symptom Variability:

The dataset includes a wide spectrum of symptom patterns ranging from mild to severe conditions. This covers short-term fatigue, chronic weakness, fluctuating mood changes, sleep disturbances, and cognitive decline. Such variability supports the Early Vitamin Deficiency Warning System by enabling it to detect both early-stage and advanced deficiency patterns.

Dietary Pattern Diversity:

Different dietary behaviors are considered, including vegetarian, vegan, mixed diets, and nutrient-deficient eating habits. Irregular meal timing and inconsistent nutrient intake are also included. This directly supports the Personalized Food Recommendation System by allowing it to generate accurate and context-aware dietary interventions based on individual eating patterns.

Medication and Drug Interaction Scenarios:

The dataset incorporates long-term medication usage patterns, particularly drugs known to affect vitamin absorption. For example, metformin is associated with Vitamin B12 deficiency risk, while proton pump inhibitors (PPIs) can impact magnesium and B12 levels. These scenarios are essential for both the Polypharmacy Risk Score System and the Vitamin Deficiency Warning System, as they enable the model to learn drug–nutrient interaction effects.

Lifestyle Variations:

Lifestyle differences such as high versus low sunlight exposure, sedentary versus active behavior, and varying stress levels are included in the dataset. These factors are critical for both the Vitamin Deficiency Prediction module and the Mental Health and Lifestyle Advisory System, as they influence vitamin synthesis, mental well-

being, and overall health outcomes.

Disease Progression Stages:

The dataset captures multiple stages of deficiency development, starting from early subtle symptoms to advanced conditions with multiple co-occurring signs. This supports early detection capabilities and enhances the system's ability to generate timely alerts before conditions worsen.

Population Diversity:

Different demographic groups are included, such as young adults, middle-aged individuals, and elderly patients. Each group has distinct metabolic rates, absorption efficiency, and risk profiles. This diversity is essential for all four components, ensuring that the system remains adaptable and accurate across varying population segments.

Overall, this structured scenario coverage ensures that all four modules of the system—vitamin deficiency detection, polypharmacy risk scoring, mental health advisory, and food recommendation—are trained on comprehensive, realistic, and diverse healthcare conditions, improving both accuracy and real-world applicability.

Data structuring and storage:

Each patient record is transformed into a standardized feature schema prior to storage to ensure consistency and compatibility across all four system components: vitamin deficiency prediction, polypharmacy risk scoring, mental health advisory, and food recommendation.

All symptom-related inputs are converted into numerical severity scores using a defined scale, enabling the Early Vitamin Deficiency Warning System to process symptom intensity effectively. Dietary intake information is represented as frequency-based nutrient indicators, reflecting the regularity of vitamin-rich food consumption. Medication history is encoded using categorical embeddings based on drug classes to support analysis in the Polypharmacy Risk Score System, particularly for identifying drug–drug and drug–nutrient interactions. Lifestyle-related factors such as sunlight exposure, physical activity, stress levels, and sleep patterns are normalized into risk-impact scores to support both mental health evaluation and vitamin deficiency prediction.

The final structured dataset is stored in a machine-readable format such as CSV, JSON, or relational database tables. This ensures efficient access for model training, evaluation, and integration across system modules while maintaining interpretability for healthcare-related analysis.

Expert validation:

To ensure clinical reliability, all synthetic mappings and rule-based transformations are

validated using domain knowledge from healthcare literature and expert guidelines. Medical plausibility is maintained by defining clinically meaningful relationships between symptoms, medications, and vitamin deficiencies. For example, combinations such as chronic fatigue with poor dietary intake and metformin usage are associated with an increased risk of Vitamin B12 deficiency. Similarly, low sunlight exposure combined with limited dietary intake indicates a higher likelihood of Vitamin D deficiency, while mood changes and sleep disturbances with folate-poor diets suggest potential folate deficiency.

Any medically unrealistic or inconsistent combinations are reviewed and corrected or removed during dataset refinement to ensure the integrity of the training data.

Dataset hygiene and quality control:

To maintain high data quality, strict validation and cleaning procedures are implemented throughout the dataset preparation process. A standardized feature format is enforced across all records to ensure uniformity. Incomplete, inconsistent, or contradictory entries are removed to prevent noise in model training. Missing values are handled using medically informed imputation strategies to preserve data reliability.

Outlier detection techniques are applied to identify abnormal patterns in symptom severity, dietary intake, and medication usage. In addition, continuous monitoring of dataset distribution is performed to prevent class imbalance, especially across different vitamin deficiency categories.

Records that fail validation checks are discarded rather than corrected arbitrarily, ensuring that only high-quality and clinically meaningful data is retained.

Normalization policy:

All input features are normalized prior to model training to ensure numerical stability while maintaining interpretability. Symptom severity scores are scaled within a defined range, typically between 0 and 1 or 0 and 5. Dietary intake is normalized based on weekly frequency of consumption. Medication duration is standardized into consistent time units such as months or years. Lifestyle factors are transformed into risk-indexed values to reflect their relative impact on health outcomes.

Despite normalization, raw data values are preserved in system logs to ensure traceability and allow clinicians or users to interpret results in a transparent and explainable manner.

Overall, this structured approach to data storage, validation, and normalization ensures that the dataset remains robust, clinically meaningful, and suitable for machine learning-based analysis across all components of the proposed system.

2.1.2. Functional, Non-Functional Requirements

Functional Requirements

1. Patient data collection and symptom logging

1.1 The system shall collect patient-reported data at regular intervals through a digital interface, including symptoms, diet, lifestyle habits, and medication history.

1.2 The system shall allow structured input of symptoms such as fatigue, weakness, mood changes, sleep disturbances, and cognitive issues using standardized severity scales.

1.3 The system shall store all patient inputs in a structured format for further processing and analysis.

Acceptance: Each patient entry is successfully recorded with complete feature fields or flagged if incomplete.

2. Data preprocessing and standardization

2.1 The system shall convert raw patient inputs into a standardized feature vector suitable for machine learning processing.

2.2 The system shall normalize symptom severity, dietary frequency, medication duration, and lifestyle indicators into consistent numeric scales.

2.3 The system shall handle missing or inconsistent values using predefined imputation rules or rejection criteria.

Acceptance : Missing or inconsistent values are properly handled using predefined rules without affecting system stability.

1. Deficiency Risk Classification (Mode-Based)

1.1 The system shall classify each case into deficiency risk categories (low,

moderate, high) based on model output thresholds.

1.2 The system shall update classification dynamically with each new patient data submission without relying on outdated states.

Acceptance Criteria: Each update produces a refreshed risk classification for each vitamin type.

2. Duration-style risk progression estimation

4.1 The system shall estimate a risk progression indicator representing the development duration of symptoms based on medication history and symptom trends.

4.2 The system shall convert this progression indicator into a normalized feature for downstream prediction models.

Acceptance: A valid progression value is generated for each patient profile update..

6. Personalized recommendation generation

6.1 The system shall generate personalized dietary recommendations based on detected vitamin deficiency risks.

6.2 The system shall provide supplementation suggestions and lifestyle modifications tailored to age, health status, and medication interactions.

6.3 The system shall ensure recommendations remain medically safe and avoid conflicts with existing medications.

Acceptance Criteria: Each moderate or high-risk case produces at least one actionable recommendation set.

7. Alert and notification system

7.1 The system shall generate alerts when patient risk levels exceed predefined thresholds for any vitamin deficiency.

7.2 The system shall notify users in real time when high-risk patterns are detected.

7.3 The system shall escalate repeated or severe risk cases as priority alerts recommending medical consultation.

Acceptance Criteria: High-risk conditions trigger notifications within the same processing cycle.

8. Longitudinal monitoring and tracking

8.1 The system shall continuously track patient symptoms, dietary habits, and medication usage over time.

8.2 The system shall update risk predictions dynamically based on newly entered historical data.

8.3 The system shall maintain a timeline of patient health profiles for trend analysis.

Acceptance Criteria: System retains multi-session history and updates risk

trends per entry.

9. Clinical decision support output

9.1 The system shall generate structured summary reports of patient risk profiles for healthcare providers.

9.2 The system shall include symptom trends, dietary gaps, medication interactions, and predicted deficiency risks in reports.

9.3 The system shall present outputs in an interpretable format suitable for clinical review.

Acceptance Criteria: Each patient profile can be exported as a structured clinical summary.

10. Safety, validation, and fallback handling

10.1 The system shall validate all inputs to ensure they are within medically plausible ranges.

10.2 The system shall reject or flag inconsistent or incomplete patient entries instead of forcing predictions.

10.3 If AI prediction is unavailable, the system shall use a rule-based fallback mechanism.

Acceptance Criteria: No invalid or corrupted data is used for final prediction output.

11. System performance and real-time operation

11.1 The system shall process patient input and generate risk predictions in real time.

11.2 The system shall ensure inference latency remains suitable for interactive clinical or mobile usage.

Acceptance Criteria: Prediction cycle completes within acceptable response time.

12. Privacy and data security compliance

12.1 The system shall securely store all patient data using encryption mechanisms.

12.2 The system shall ensure that only authorized users can access patient records and clinical reports.

12.3 The system shall maintain data integrity across all processing stages.

Acceptance Criteria: All patient data remains protected and accessible only through authenticated access layers.

3. Vitamin deficiency risk analysis (AI model)

3.1 The system shall analyze processed patient data using machine learning models to estimate the risk of vitamin deficiencies such as B12, D, and folate.

3.2 The system shall detect patterns between symptoms, diet, lifestyle, and medication usage that indicate potential deficiencies.

3.3 The system shall output a risk score and severity level for each suspected vitamin deficiency.

Acceptance: Risk prediction is generated for each vitamin category with a numeric confidence score.

Non-Functional Requirements

1. Accuracy and effectiveness

The system shall accurately detect vitamin deficiency risks based on patient symptoms, dietary intake, lifestyle factors, and medication history, ensuring minimal false positives and false negatives. It should provide consistent and reliable risk scores along with personalized recommendations across different users and health conditions.

2. Reliability and robustness

The system shall maintain stable operation even when dealing with missing, incomplete, or noisy input data. It shall apply fallback mechanisms such as rule-based estimation when necessary and recover automatically when complete data becomes available, ensuring uninterrupted functionality

3. Safety and compliance

The system shall follow established medical and nutritional guidelines when generating recommendations. It should ensure that all outputs are safe, non-invasive, and intended for general health support only, without providing direct medical diagnosis or treatment instructions.

4. Portability and deployment

The system should be designed to operate across multiple platforms, including web and mobile environments, with minimal computational requirements. The trained machine learning models should be optimized for deployment on standard devices and scalable for future enhancements or increased user demand.

5. Explainability

The system shall provide transparent and interpretable outputs by linking predictions to input features such as symptoms, dietary habits, and medication history. It shall ensure that users and healthcare providers can trace how each risk prediction and recommendation is generated, improving trust and usability in clinical decision support.

2.2. Commercialization Aspects of the Product

The *Smart Polypharmacy and Vitamin Recommendation System* has strong potential for commercialization due to the increasing global demand for intelligent healthcare solutions, particularly in elderly care and chronic disease management. With the rise in polypharmacy cases and vitamin deficiency-related complications, there is a growing need for affordable, scalable, and AI-driven decision-support systems that can assist both patients and healthcare providers.

From a market perspective, the system can be positioned as a digital health platform targeting hospitals, clinics, pharmacies, and home-care service providers. It can also be extended as a mobile and web-based application for individual users, allowing

elderly patients and caregivers to monitor health risks, receive dietary recommendations, and track medication-related effects in real time. This dual-market approach (B2B and B2C) increases its commercial viability and scalability.

The system's modular architecture provides additional business value, as each component can be commercialized separately or as an integrated solution. For example, the Polypharmacy Risk Score System can be offered as a clinical decision-support tool for healthcare institutions, while the Food Recommendation System can be adapted for nutrition and wellness applications. Similarly, the Mental Health Advisory module can be integrated into elderly care platforms and telehealth services. Another key advantage of commercialization is the use of artificial intelligence and machine learning, which enables automation and reduces the dependency on manual clinical assessments. This improves efficiency for healthcare providers and reduces operational costs. Additionally, the system can be offered under a Software-as-a-Service (SaaS) model, allowing subscription-based access for hospitals, pharmacies, and individual users.

Scalability is also a major factor supporting commercialization. The system can be expanded to support multiple regions, languages, and dietary cultures, making it suitable for global deployment. Furthermore, integration with wearable devices and electronic health records (EHR) systems can enhance its value proposition and open opportunities for partnerships with healthcare technology providers.

Overall, the proposed system has strong commercial potential due to its practical healthcare applications, modular design, and adaptability to both clinical and consumer markets, making it a viable solution in the growing digital health industry.

2.3. Testing and Implementations

2.3.1. Implementations

2.3.1.1. AI Based Vitamin deficiency Risk Environment Development

The AI-based vitamin deficiency risk environment is the core analytical component of the proposed *Smart Polypharmacy and Vitamin Recommendation System*. It is designed to process multi-dimensional patient data and predict the likelihood of vitamin deficiencies such as Vitamin B12, Vitamin D, and folate. This module integrates machine learning techniques with structured healthcare data to provide early warning insights and personalized health recommendations.

The development of this environment begins with the construction of a unified data pipeline that combines symptoms, dietary patterns, lifestyle factors, and long-term medication history. Each input is preprocessed and transformed into a standardized numerical format to ensure compatibility with machine learning models. Feature engineering techniques are applied to capture meaningful relationships between variables, such as the impact of specific medications on nutrient absorption and the correlation between lifestyle habits and deficiency risk.

Several machine learning algorithms are evaluated during the implementation phase, including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes. These models are trained on the processed dataset to classify vitamin deficiency risk levels. Performance comparison is conducted using evaluation metrics such as accuracy, precision, recall, and F1-score to select the most effective model for deployment.

The selected model is then integrated into the system as the primary prediction engine. It generates a risk score for each vitamin deficiency category based on real-time patient input. The output is categorized into risk levels such as low, moderate, and high, enabling early detection and timely intervention. Additionally, the system incorporates a progression-based feature that estimates the severity and development stage of potential deficiencies over time.

To enhance interpretability, the AI environment is designed with explainable outputs, allowing users and healthcare providers to understand how specific symptoms, dietary habits, and medication patterns contribute to the final prediction. This transparency improves trust and usability in clinical decision-making.

Furthermore, the system is optimized for real-time processing, ensuring that predictions are generated efficiently upon each user input. The environment is designed to be scalable, allowing future integration of additional data sources such as wearable devices and electronic health records (EHRs).

Overall, the AI-based vitamin deficiency risk environment serves as the foundation of the proposed system by enabling intelligent, data-driven, and proactive healthcare decision support.

2.3.1.2. Decision-Making Function Development

The decision-making function is a critical component of the proposed *Smart Polypharmacy and Vitamin Recommendation System*, responsible for converting

processed patient data and AI model outputs into meaningful clinical insights and actionable recommendations. This module acts as the bridge between raw predictive results and user-understandable health guidance.

The development of the decision-making function is based on a rule-enhanced machine learning framework, where model predictions are combined with domain knowledge from medical guidelines. The function takes inputs such as vitamin deficiency risk scores, polypharmacy risk levels, symptom severity, dietary patterns, and medication history to generate final decisions regarding patient health status.

A threshold-based classification mechanism is implemented to categorize risk levels into low, moderate, and high. These thresholds are dynamically adjustable to ensure flexibility across different patient profiles and clinical scenarios. The system prioritizes high-risk cases by triggering immediate recommendations and alerts, while moderate cases are handled through preventive guidance and monitoring suggestions. In addition to risk classification, the decision-making function incorporates drug–nutrient interaction rules to ensure safe and medically valid outputs. For example, if a patient is identified with Vitamin B12 deficiency and is using metformin, the system prioritizes dietary recommendations rich in B12 sources while considering medication constraints. Similarly, lifestyle factors such as low sunlight exposure are factored into Vitamin D-related recommendations.

The module also supports multi-factor decision fusion, where multiple inputs (symptoms, diet, medication, and lifestyle) are weighted and combined to produce a final decision score. This ensures that no single factor dominates the outcome, resulting in a more balanced and realistic assessment of patient health conditions.

To improve transparency and usability, the decision-making function is designed with explainability features. Each output is accompanied by a justification that highlights the contributing factors behind the recommendation or risk score. This allows healthcare providers and users to understand how decisions are derived, increasing trust in the system.

Overall, the decision-making function enhances the intelligence of the system by ensuring that predictions are not only accurate but also clinically meaningful, safe, and interpretable for real-world healthcare applications.

2.3.1.3. AI Model development (Risk Prediction System)

The AI Model Development component forms the core predictive engine of the proposed *Smart Polypharmacy and Vitamin Recommendation System*. This module is responsible for analyzing patient data and generating risk predictions for vitamin deficiencies and polypharmacy-related complications using machine learning techniques.

The development process begins with data preparation, where structured patient inputs such as symptoms, dietary intake, medication history, and lifestyle factors are transformed into feature vectors suitable for model training. Feature engineering techniques are applied to enhance predictive performance by capturing meaningful relationships between variables, such as drug–nutrient interactions and symptom progression patterns.

Multiple machine learning algorithms are evaluated during the model development phase, including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes. These models are trained to classify vitamin deficiency risk levels and identify potential high-risk polypharmacy cases. The performance of each model is assessed using evaluation metrics such as accuracy, precision, recall, and F1-score to ensure reliability and consistency in medical prediction tasks.

To address the complexity of real-world healthcare scenarios, the system adopts a multi-label prediction approach, allowing simultaneous detection of multiple vitamin deficiencies. This is particularly important as patients under polypharmacy conditions may experience more than one deficiency at the same time. The model is further enhanced using probability-based scoring to represent the severity and confidence of each prediction.

The selected optimized model is integrated into the system as the main risk prediction engine. It processes real-time patient data and outputs a risk score for each vitamin deficiency category along with a corresponding severity level (low, moderate, or high). This enables early detection and supports preventive healthcare decision-making.

In addition, a semi-supervised learning approach is incorporated to improve model robustness in situations where labeled medical data is limited. By leveraging both labeled and unlabeled data, the system enhances generalization and reduces dependency on extensive manual annotation.

Explainability is also a key aspect of the AI model. The system provides interpretable outputs by identifying which features (such as specific symptoms, medications, or dietary patterns) contributed most to the prediction. This ensures transparency and improves trust among healthcare professionals and end users.

Overall, the AI model development phase establishes a robust, scalable, and intelligent risk prediction system capable of supporting early diagnosis, personalized recommendations, and improved healthcare outcomes in polypharmacy and vitamin deficiency management.

2.3.1.4. Multi-Agent Approach

The Multi-Agent Approach is introduced in the proposed *Smart Polypharmacy and Vitamin Recommendation System* to improve modularity, scalability, and decision accuracy across different healthcare tasks. Instead of relying on a single monolithic model, the system is divided into multiple intelligent agents, where each agent is responsible for a specific functional domain such as vitamin deficiency detection, polypharmacy risk analysis, mental health advisory, and food recommendation generation.

Each agent operates independently while sharing relevant information through a centralized data exchange layer. This structure enables parallel processing of patient data and ensures that each health aspect is analyzed using specialized logic and models. For example, the Vitamin Deficiency Agent focuses on symptom-diet relationships, while the Polypharmacy Agent evaluates drug–drug and drug–nutrient interactions.

The interaction between agents is managed through a coordination mechanism that integrates output into a unified decision framework. This ensures that final recommendations are not isolated but instead reflect a holistic view of the patient’s condition. For instance, if the Polypharmacy Agent detects a high-risk drug interaction, the Food Recommendation Agent adjusts dietary suggestions to avoid conflicting nutrients.

The Multi-Agent System also enhances adaptability by allowing each agent to be updated or improved independently without affecting the entire system. This makes the architecture more flexible for future enhancements, such as adding new disease modules or integrating wearable sensor data.

Communication between agents is designed using a structured message-passing approach, where standardized data formats are used to ensure consistency. Each agent receives processed input features and returns structured outputs such as risk scores, alerts, or recommendations, which are then aggregated by the decision-making layer.

Another key advantage of the multi-agent design is improved fault tolerance. If one agent fails or produces uncertain outputs, the remaining agents can continue functioning, ensuring system reliability. This is particularly important in healthcare applications where continuous availability is critical.

Overall, the Multi-Agent Approach strengthens the proposed system by enabling distributed intelligence, improving prediction accuracy, and ensuring scalable and modular system architecture suitable for complex healthcare decision support environments.

2.3.1.5. Risk Aggregation and Recommendation Output

The Risk Aggregation and Recommendation Output module is responsible for combining output from all AI components and transforming them into a unified, clinically meaningful decision support output. This module plays a key role in ensuring that the results generated by the Vitamin Deficiency Risk System, Polypharmacy Risk System, Mental Health Advisory System, and Food Recommendation System are consolidated into a single coherent patient profile.

The aggregation process begins by collecting risk scores generated from different predictive models. Each output is assigned a weighted importance based on its clinical relevance. For example, drug–nutrient interaction risks from the polypharmacy module are given higher priority when they directly influence vitamin absorption. Similarly, symptom-based deficiency predictions are combined with dietary and lifestyle risk indicators to produce a final composite risk score for each vitamin category.

Once aggregation is completed, the system classifies the overall patient condition into predefined severity levels such as low, moderate, and high risk. This classification ensures that healthcare providers and users can quickly interpret the patient’s health status without needing to analyze individual model outputs separately.

The recommendation generation process is directly linked to the aggregated risk results. For high-risk cases, the system generates immediate intervention strategies, including dietary modifications, supplementation suggestions, and medical consultation alerts. For moderate-risk cases, preventive guidance and lifestyle adjustments are provided, while low-risk cases receive maintenance-level recommendations to sustain healthy conditions.

A key feature of this module is its context-aware recommendation engine, which ensures that all suggestions are safe and personalized. Drug–nutrient interactions are carefully considered to avoid harmful combinations. For instance, if a patient is on long-term metformin therapy, the system prioritizes Vitamin B12-rich food recommendations while ensuring compatibility with existing medications and dietary restrictions.

The output is presented in a structured and interpretable format, including risk scores, contributing factors, and recommended actions. This enhances transparency and supports clinical decision-making by allowing healthcare professionals to understand how each recommendation is derived.

Additionally, the module supports real-time updates, meaning that recommendations are dynamically adjusted whenever new patient data is received. This ensures continuous monitoring and timely intervention in rapidly changing health conditions.

Overall, the Risk Aggregation and Recommendation Output module integrates multiple intelligent subsystems into a unified decision framework, enabling

accurate, safe, and personalized healthcare guidance for patients under polypharmacy and vitamin deficiency risk conditions.

2.3.2. Testing

Testing is a critical phase in the development of the *Smart Polypharmacy and Vitamin Recommendation System*, ensuring that all modules function correctly, efficiently, and reliably under different scenarios. The testing process is designed to validate both functional and non-functional requirements, covering system accuracy, robustness, safety, and performance across all four main components of the system.

The testing strategy includes multiple levels, starting from unit testing, where individual modules such as data preprocessing, risk prediction, and recommendation generation are tested independently. This ensures that each component performs as expected before integration. Following this, integration testing is conducted to verify that all modules—vitamin deficiency detection, polypharmacy risk scoring, mental health advisory, and food recommendation systems—work together seamlessly and exchange data correctly.

System testing is performed to evaluate the complete end-to-end workflow, from patient data input to final recommendation output. This phase ensures that the system can handle real-world scenarios, including incomplete inputs, noisy data, and multiple concurrent risk conditions. Special attention is given to verifying the correctness of multi-label predictions, where a patient may be identified with more than one vitamin deficiency simultaneously.

To evaluate model performance, standard machine learning evaluation metrics such as accuracy, precision, recall, and F1-score are used. These metrics help assess the reliability of the AI-based prediction models used in vitamin deficiency detection and polypharmacy risk scoring. Confusion matrices are also analyzed to understand misclassification patterns and improve model performance.

User acceptance testing (UAT) is conducted with potential end-users such as healthcare providers and caregivers to evaluate the usability and practicality of the system. Feedback from this phase is used to improve the clarity of outputs, recommendation relevance, and interface design.

In addition, stress testing is performed to evaluate system performance under high data loads and multiple simultaneous user requests. This ensures that the system maintains acceptable response times and stability in real-time healthcare environments.

Security testing is also included to verify that patient data is securely stored and accessed only by authorized users. This ensures compliance with data privacy and healthcare data protection standards.

Overall, the testing phase confirms that the system is accurate, reliable, scalable, and safe for deployment in real-world healthcare settings, ensuring that it can effectively support early vitamin deficiency detection and polypharmacy risk management.

2.3.2.1. Test objectives and scope

The primary objective of testing the *Smart Polypharmacy and Vitamin Recommendation System* is to ensure that the system operates accurately, reliably, and safely in predicting vitamin deficiency risks and polypharmacy-related complications. Testing is carried out to determine that all functional modules perform as intended and that the integrated system delivers consistent and meaningful healthcare insights.

One of the key objectives is to verify the correctness of the AI-based prediction models used in vitamin deficiency detection and risk scoring. This includes ensuring that the models correctly analyze symptom data, dietary patterns, lifestyle factors, and medication history to generate accurate risk classifications. Another objective is to validate the decision-making and recommendation modules, ensuring that the system produces safe, relevant, and personalized outputs without conflicting with medical constraints.

The testing process also aims to evaluate system robustness under real-world conditions, including scenarios with missing, incomplete, or inconsistent data. This is important to ensure that the system can still function effectively using fallback mechanisms or rule-based logic when necessary. In addition, performance objectives focus on ensuring that the system responds in real time, making it suitable for clinical and mobile usage environments.

From a usability perspective, testing ensures that healthcare providers and users can easily interact with the system, understand outputs, and interpret risk levels without confusion. Explainability of predictions is also evaluated to confirm that the system provides transparent reasoning behind each recommendation.

- The scope of testing covers all four major components of the system:
- Vitamin Deficiency Early Warning System
 - Polypharmacy Risk Score System
 - Mental Health and Lifestyle Advisory Module
 - Personalized Food Recommendation System

In addition, testing includes data preprocessing, feature engineering, AI model performance, decision-making logic, and multi-agent integration. Both individual module testing and full system-level validation are included within the scope.

Non-functional aspects such as security, scalability, reliability, and response time are also evaluated to ensure that the system meets healthcare-grade standards. User acceptance testing is included in validating real-world applicability and gather feedback from potential end users such as healthcare professionals and caregivers.

Overall, the test objectives and scope ensure that the proposed system is accurate, safe, scalable, and practical for deployment in real healthcare environments, particularly for managing elderly patients under polypharmacy conditions.

2.3.2.2. Test environment and configuration

The test environment for the *Smart Polypharmacy and Vitamin Recommendation System* is designed to closely simulate real-world healthcare usage scenarios, ensuring that the system performs reliably under practical conditions. The environment supports both development-level validation and system-level evaluation across all four main components: vitamin deficiency detection, polypharmacy risk scoring, mental health advisory, and food recommendation modules.

The system is implemented and tested in a controlled software environment using a standard development stack that supports machine learning workflows, data preprocessing, and API-based integration. The backend environment is configured to handle data processing, model inference, and decision-making logic, while the frontend environment is used to test user interaction, data input forms, and result visualization.

For machine learning model testing, a Python-based environment is used with libraries supporting data processing, feature extraction, and classification models. This includes support for training and evaluating algorithms such as Logistic Regression, Support Vector Machines, K-Nearest Neighbors, and Naïve Bayes. The environment also supports performance evaluation using standard metrics such as accuracy, precision, recall, and F1-score.

The data set used in testing consists of both synthetic and structured patient-reported data, covering a wide range of scenarios including different symptom severities, dietary patterns, medication histories, and lifestyle conditions. This ensures that the system is tested across diverse and realistic healthcare conditions, particularly focusing on elderly patients under polypharmacy.

The hardware configuration used for testing includes a standard computing setup capable of handling moderate-scale machine learning workloads. This ensures that model training and inference can be evaluated under realistic resource constraints similar to those found in typical healthcare systems or cloud-based deployments.

The software configuration includes a web-based interface for user interaction, allowing simulated patient data entry and real-time output visualization. API integration is used to connect the frontend interface with backend prediction models, ensuring smooth data flow across system modules.

In addition, version control tools and testing frameworks are used to manage code changes and ensure consistent testing results. Logging mechanisms are implemented to track system behavior, errors, and prediction outputs for further analysis and debugging.

Overall, the test environment and configuration are designed to provide a realistic, scalable, and controlled setup that ensures the system's performance, reliability, and accuracy can be thoroughly evaluated before deployment in real-world healthcare settings.

2.3.2.3. Datasets and scenarios

The testing phase of the *Smart Polypharmacy and Vitamin Recommendation System* utilizes a combination of structured datasets and scenario-based simulations to ensure comprehensive evaluation of all system components. Since real clinical datasets with detailed symptom–medication–nutrition relationships are often limited or incomplete, a hybrid dataset approach is adopted to ensure both realism and diversity in testing conditions.

The primary dataset consists of patient-reported and synthetically generated records that simulate real-world healthcare conditions. Each record includes structured information such as symptom profiles, dietary intake patterns, lifestyle factors, demographic details, and medication history. This enables the system to be tested under various clinical situations, particularly those involving elderly patients with polypharmacy conditions.

To ensure robustness, the dataset includes multiple scenario types. Symptom-based scenarios cover mild, moderate, and severe manifestations such as fatigue, weakness, cognitive decline, and sleep disturbances. Dietary scenarios include balanced diets, vitamin-deficient diets, vegetarian and vegan patterns, and irregular eating habits. Medication-related scenarios simulate long-term drug usage such as metformin, proton pump inhibitors, and anticonvulsants, which are known to affect nutrient absorption. Lifestyle scenarios include variations in sunlight exposure, physical activity levels, and stress conditions.

In addition, progression-based scenarios are included to simulate the evolution of vitamin deficiencies over time. These scenarios allow the system to evaluate early warning capabilities by distinguishing between early-stage symptoms and advanced deficiency conditions. Multi-condition scenarios are also incorporated, where patients exhibit more than one vitamin deficiency simultaneously, reflecting real-world complexity.

Special attention is given to polypharmacy scenarios, where multiple medications are used concurrently. These scenarios are designed to test the system’s ability to detect drug–drug and drug–nutrient interactions and assess overall risk levels accurately. This is critical for validating the Polypharmacy Risk Score System component.

The dataset also includes edge cases such as incomplete data entries, missing symptom reports, and inconsistent medication histories. These scenarios are used to evaluate the system’s robustness and ability to handle imperfect real-world inputs using imputation or fallback mechanisms.

Overall, the combination of structured datasets and diverse testing scenarios ensures that the system is thoroughly evaluated across a wide range of realistic healthcare conditions. This approach enhances the reliability, accuracy, and generalizability of the proposed solution in managing vitamin deficiencies and polypharmacy risks.

2.3.2.4. Metrics and acceptance criteria

The evaluation of the *Smart Polypharmacy and Vitamin Recommendation System* is carried out using a combination of machine learning performance metrics and system-level acceptance criteria. These measures ensure that the system not only performs well from an algorithmic perspective but also meets practical healthcare requirements in terms of reliability, safety, and usability.

For the AI-based vitamin deficiency risk prediction component, standard classification metrics are used. Accuracy is used to measure the overall correctness of predictions across all classes. Precision evaluates the proportion of correctly identified positive cases among all predicted positives, while recall measures the system's ability to identify actual deficiency cases. The F1-score is used to provide a balanced measure between precision and recall, particularly in cases of imbalanced datasets where certain deficiency classes may be underrepresented. Confusion matrix analysis is also applied to identify misclassification patterns and improve model performance.

For the polypharmacy risk scoring module, evaluation focuses on the system's ability to correctly identify high-risk drug interaction cases. Sensitivity to drug–drug and drug–nutrient interactions is assessed to ensure that the system can detect clinically significant risks. Risk stratification accuracy is measured by comparing predicted risk categories (low, moderate, high) against validated reference scenarios.

In the recommendation generation module, evaluation is based on relevance, safety, and consistency. Recommendations are assessed to ensure they align with detected deficiency risks and do not conflict with medication constraints. Expert validation is used to verify that dietary and lifestyle suggestions are medically appropriate and safe for users.

System-level performance metrics are also considered. Response time is measured to ensure that predictions and recommendations are generated in real time, suitable for clinical or mobile use. Robustness is evaluated by testing system behavior under missing or noisy data conditions. Reliability is assessed based on the system's ability to maintain consistent performance across multiple user sessions and diverse input scenarios.

Explainability is another important evaluation factor. The system is assessed on its ability to clearly justify predictions by linking outputs to input features such as symptoms, dietary habits, and medication history. This ensures transparency and improves user trust, especially in healthcare decision-making environments.

Acceptance criteria are defined to ensure the system meets minimum performance standards before deployment. The vitamin deficiency prediction model must achieve acceptable thresholds for accuracy, precision, recall, and F1-score based on validation datasets. The system must consistently generate correct risk classifications for all vitamin categories with minimal error rates. High-risk cases must trigger appropriate alerts and recommendations within the same processing cycle. The system must handle incomplete or missing data without failure using fallback mechanisms. All recommendations must pass safety validation checks and must not conflict with known drug interactions.

Additionally, the system must generate outputs within acceptable response times suitable for real-time usage. User-facing outputs must be interpretable and consistent across repeated tests. Finally, the system must maintain data integrity and ensure secure handling of all patient-related information.

Overall, these metrics and acceptance criteria ensure that the proposed system is accurate, safe, reliable, and suitable for real-world healthcare deployment.

2.3.2.5. Module (unit) testing

Module or unit testing is conducted to verify the correctness and functionality of individual components within the *Smart Polypharmacy and Vitamin Recommendation System*. Each module is tested independently to ensure that it performs its intended function accurately before integration into the full system. This approach helps in early detection of errors and improves overall system reliability.

The unit testing process begins with the **data preprocessing module**, where patient inputs such as symptoms, dietary information, medication history, and lifestyle factors are validated. Tests ensure that raw inputs are correctly transformed into standardized numerical formats, missing values are handled appropriately, and normalization rules are correctly applied.

The **vitamin deficiency prediction module** is tested by providing controlled input cases with known expected outcomes. This verifies whether the machine learning model correctly identifies risk levels for deficiencies such as Vitamin B12, Vitamin D, and folate. The output is evaluated based on consistency, classification accuracy, and stability across repeated inputs.

The **polypharmacy risk scoring module** is tested by simulating different medication combinations, especially those known to cause drug–drug or drug–nutrient interactions. The objective is to confirm that the system correctly assigns risk levels based on interaction severity and patient-specific factors.

The **recommendation generation module** is tested by checking whether the system produces safe and relevant dietary and lifestyle suggestions based on predicted deficiencies. Each output is verified against predefined medical rules to ensure that recommendations do not conflict with known drug interactions or nutritional guidelines.

The **decision-making and aggregation module** is tested to ensure that multiple inputs (symptoms, diet, medication, and lifestyle) are correctly combined to produce a final risk score and classification. This includes verifying weighting mechanisms and ensuring consistent output generation across similar input cases.

Additionally, the **alert and notification module** is tested by simulating high-risk scenarios to ensure that alerts are triggered correctly and in real time. The system's fallback mechanisms are also tested by introducing missing or incomplete data to verify graceful degradation without system failure.

Overall, module testing ensures that each individual component functions correctly in isolation, providing a stable foundation for integration testing and full system deployment. This step is critical in ensuring the accuracy, safety, and reliability of the proposed healthcare system.

2.3.2.6. Risk-model verification

Risk-model verification is performed to ensure that the predictive models used in the *Smart Polypharmacy and Vitamin Recommendation System* produce accurate, stable, and clinically meaningful risk assessments. This phase focuses specifically on validating the correctness of the vitamin deficiency risk prediction models and the polypharmacy risk scoring mechanism under different healthcare scenarios.

The verification process begins by evaluating the consistency of model outputs across repeated tests using the same input data. This ensures that the model produces stable risk predictions without significant variation, which is essential for reliability in healthcare decision-making. Any inconsistencies are further analyzed to identify potential issues in feature processing, model training, or threshold configuration.

Next, the models are verified using known benchmark scenarios that represent typical clinical conditions. These include cases of long-term medication use such as metformin-related Vitamin B12 deficiency risk, low sunlight exposure associated with Vitamin D deficiency, and poor dietary intake leading to folate deficiency. The expected outputs for these scenarios are compared against actual model predictions to validate clinical correctness.

The verification also includes cross-validation techniques to assess how well the model generalizes to unseen data. This helps ensure that the system does not overfit to training data and can maintain reliable performance across different patient profiles. Performance metrics such as accuracy, precision, recall, and F1-score are analyzed during this process to confirm model effectiveness.

In addition, threshold validation is performed to ensure that risk categories (low, moderate, high) are appropriately defined and aligned with medical relevance. This involves fine-tuning decision boundaries to reduce false positives and false negatives, especially in high-risk cases where early detection is critical.

The polypharmacy risk model is further verified by testing drug–drug and drug–nutrient interaction scenarios. The system is evaluated on its ability to correctly identify high-risk medication combinations and assign appropriate risk levels based on interaction severity and patient-specific factors.

Explainability verification is also conducted to ensure that the model can clearly justify its predictions. Each risk output is checked to confirm that it is traceable to input features such as symptoms, dietary patterns, or medication history. This is essential for building trust in clinical environments where transparency is critical.

Overall, risk-model verification ensures that the AI components of the system are not only statistically accurate but also clinically valid, stable, and interpretable. This step is essential for confirming that the system is safe and reliable for real-world healthcare applications involving elderly patients and polypharmacy conditions.

2.3.2.7. Multi-agent selection tests

Multi-agent selection tests are conducted to evaluate the correctness, coordination, and efficiency of the Multi-Agent System within the *Smart Polypharmacy and Vitamin Recommendation System*. This testing phase ensures that the appropriate agent is selected for each input scenario and that all agents work collaboratively to produce a unified and accurate healthcare decision.

The testing process begins by validating the **agent selection mechanism**, which is responsible for routing incoming patient data to the correct specialized agent. For example, symptom-dominant inputs are directed to the Vitamin Deficiency Agent, medication-heavy inputs are routed to the Polypharmacy Risk Agent, and lifestyle or emotional data is directed to the Mental Health Advisory Agent. Test cases are designed to verify that each input type is correctly classified and assigned to the appropriate processing module.

Next, **inter-agent communication tests** are performed to ensure that agents can effectively exchange relevant information. Since the system relies on shared patient data across multiple agents, it is essential that outputs from one agent (such as risk scores or interaction alerts) are correctly interpreted and utilized by other agents. These tests confirm that data consistency is maintained across the system.

The system is also tested for **conflict resolution between agents**. In certain scenarios, different agents may produce overlapping or conflicting recommendations. For example, a dietary suggestion from the Food Recommendation Agent may need to be adjusted based on a warning from the Polypharmacy Risk Agent. The testing ensures that the coordination layer properly resolves such conflicts and generates a safe, unified output.

Additionally, **load distribution and parallel processing tests** are conducted to evaluate system performance when multiple agents are triggered simultaneously. This ensures that the system maintains efficiency and low response time even when handling complex multi-condition patient profiles.

Fallback behavior testing is also included to verify system stability when one or more agents fail or produce uncertain outputs. In such cases, the system should continue functioning using alternative logic or rule-based estimations without disrupting the overall decision flow.

Finally, **output consistency tests** are performed to ensure that the final aggregated result remains stable and logically coherent regardless of which combination of agents is activated. This confirms that the multi-agent architecture produces reliable and reproducible healthcare recommendations.

Overall, multi-agent selection testing validates that the system correctly assigns tasks to specialized agents, maintains smooth inter-agent communication, resolves conflicts effectively, and produces a unified, accurate, and safe healthcare decision output.

2.3.2.8. Logging, observability, and traceability

Logging, observability, and traceability are implemented as essential testing and monitoring mechanisms within the *Smart Polypharmacy and Vitamin Recommendation System* to ensure transparency, debugging capability, and system reliability. These mechanisms support continuous evaluation of system behavior across all modules, including vitamin deficiency prediction, polypharmacy risk scoring, mental health advisory, and food recommendation generation.

The **logging system** records all critical system events, including patient data inputs, preprocessing steps, model predictions, decision outputs, and recommendation generation. Each log entry includes timestamps, module identifiers, input features, and output results. This allows developers to track the full lifecycle of each patient interaction and identify any inconsistencies or errors during processing.

Observability is incorporated to provide real-time insight into system performance and behavior. Key performance indicators such as prediction latency, model confidence scores, data processing time, and agent execution flow are continuously monitored. This ensures that the system operates efficiently and helps identify performance bottlenecks during high-load scenarios or complex multi-agent interactions.

The **traceability mechanism** ensures that every output generated by the system can be traced back to its original input data. This is particularly important in healthcare applications where explainability and accountability are critical. For example, a vitamin deficiency risk prediction can be traced back to specific symptoms, dietary patterns, and medication history that contributed to the final result.

In addition, traceability supports the explainable AI requirement by providing a clear audit trail for each decision. Healthcare providers can review how a particular recommendation was generated, which features had the highest influence, and which model components were involved in the decision-making process.

Error logging and exception tracking are also implemented to capture system failures, invalid inputs, and unexpected model behaviors. These logs are used to refine system performance, improve data preprocessing rules, and enhance model robustness over time.

Furthermore, observability tools enable system administrators to monitor resource usage, response times, and system health in real time. This ensures that the system remains stable and scalable when deployed in real-world healthcare environments.

Overall, logging, observability, and traceability collectively enhance system transparency, reliability, and maintainability. They ensure that the proposed system not only performs accurately but also remains explainable, auditable, and trustworthy for clinical and research applications.

3. Results & Discussion

3.1. Results

This section presents quantitative and qualitative evidence demonstrating that the proposed *Smart Polypharmacy and Vitamin Recommendation System* effectively identify early vitamin deficiency risks using patient-reported symptoms, dietary patterns, lifestyle factors, and long-term medication history. The evaluation is conducted using a standardized pipeline that includes both real-world inspired and synthetically generated patient profiles to ensure coverage of diverse clinical scenarios

Evaluation metrics

The system performance is assessed using clinically relevant and machine learning-based evaluation metrics to ensure both predictive accuracy and healthcare applicability:

- **Prediction Accuracy (%)**: Measures the correctness of vitamin deficiency classification (Vitamin B12, Vitamin D, folate) across low, moderate, and high risk levels.
- **Risk Detection Sensitivity (%)**: Evaluates the ability to correctly identify true deficiency cases, particularly in early-stage conditions.
- **False Negative Rate (%)**: Represents missed high-risk cases, which is a critical safety indicator in healthcare applications.
- **Risk Score Error**: Measures deviation between predicted risk scores and expected reference values.
- **Response Consistency**: Assesses stability of predictions for repeated inputs of the same patient profile.
- **Recommendation Relevance Score**: Evaluates how well generated dietary and lifestyle recommendations align with detected deficiency conditions.

These metrics are computed across multiple patient scenarios, including medication-heavy profiles, nutrient-deficient diets, and lifestyle-imbalanced cases.

Overall performance

The system demonstrates strong and stable performance across diverse patient conditions. In most evaluation cases, the model successfully identifies vitamin deficiency risks with high consistency and reliability.

- Mild deficiency cases are accurately classified as low risk, with minimal variation in repeated predictions.
- Moderate cases are consistently mapped to medium risk levels, reflecting early warning capability.
- Severe deficiency profiles are correctly identified as high risk, with appropriate alert generation and recommendation triggering.

Although minor misclassifications occur in borderline cases where input data is incomplete or ambiguous, the system improves accuracy over time through longitudinal tracking and iterative updates.

Progressive design evaluation

The system's performance improves significantly through successive design enhancements:

- **Structured Feature Representation:** Conversion of raw inputs into normalized feature vectors improves stability and reduces noise sensitivity.
- **Multi-Source Integration:** Combining symptoms, dietary intake, lifestyle, and medication history increases prediction accuracy compared to single-source models.
- **Multi-Agent Modeling:** Separation of tasks across specialized agents improves sensitivity for both early and advanced deficiency detection.
- **Progression Tracking Mechanism:** Temporal analysis of symptom changes enhances early warning capability and reduces delayed detection.
- **Recommendation Filtering Layer:** Safety constraints improve clinical reliability and eliminate contradictory or unsafe outputs.

These enhancements collectively result in improved robustness, higher prediction accuracy, and better clinical alignment.

Learning and prediction behavior

The system demonstrates adaptive learning behavior as more patient data becomes available over time. Initial predictions may show slight uncertainty; however, accuracy improves significantly once medication history and symptom progression are incorporated.

The multi-agent architecture ensures smoother transitions between risk levels, reducing abrupt fluctuations in classification. This results in more stable and clinically meaningful predictions, particularly for long-term monitoring scenarios.

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Risk detection and safety performance

The system successfully identifies clinically relevant deficiency patterns, including:

- Chronic fatigue combined with poor dietary intake → **Vitamin B12 deficiency risk**
- Low sunlight exposure → **Vitamin D deficiency risk**
- Long-term use of absorption-affecting medications → **increased deficiency likelihood**

Safety mechanisms ensure that all recommendations remain medically appropriate. The system avoids unsafe supplementation suggestions and enforces clinically valid thresholds for alert generation.

Longitudinal stability and tracking

Continuous monitoring results show that the system maintains stable risk trends over time. Increasing symptom severity leads to gradual risk score escalation, while improvements in lifestyle or diet reflect corresponding risk reduction.

This confirms that the system supports **longitudinal health tracking**, rather than static one-time prediction, making it suitable for real-world chronic condition monitoring.

Summary

- Structured symptom–diet–medication integration significantly improves early detection accuracy
- Multi-agent architecture enhances sensitivity across all deficiency severity levels
- Longitudinal tracking improves prediction stability and reduces misclassification
- Safety constraints ensure medically consistent and non-harmful

recommendations

- The system reliably detects early risks for Vitamin B12, Vitamin D, and folate deficiencies

Overall, the experimental results confirm that the proposed system successfully achieves its primary objective of early, accurate, and clinically meaningful detection of vitamin deficiency risks. In addition, it provides safe, personalized, and explainable recommendations, making it suitable for real-world application in elderly healthcare and polypharmacy management scenarios.

3.2. Research Findings

The findings of this research highlight the effectiveness of combining machine learning, multi-agent systems, and rule-based clinical reasoning for early detection and management of vitamin deficiency risks in elderly patients, particularly those affected by polypharmacy. The study demonstrates that integrating symptom data, dietary behavior, lifestyle factors, and medication history provides a more comprehensive and accurate representation of patient health compared to traditional single-source diagnostic approaches.

A key finding is that **early vitamin deficiency risk can be reliably predicted before severe clinical symptoms appear**. The system was able to detect patterns associated with Vitamin B12, Vitamin D, and folate deficiencies using indirect indicators such as fatigue, sleep disturbance, mood changes, and long-term medication usage. This confirms that non-laboratory, data-driven approaches can support early warning systems in preventive healthcare.

Another important finding is that **medication history plays a significant role in vitamin deficiency risk prediction**. Long-term use of drugs such as metformin and acid-reducing agents was strongly associated with increased risk of Vitamin B12 and nutrient absorption issues. This supports existing medical literature that emphasizes drug–nutrient interactions as a major but often under-monitored factor in elderly healthcare.

The research also found that **multi-source data integration significantly improves prediction reliability**. Models that included only symptoms showed lower accuracy compared to models that incorporated diet, lifestyle, and medication history together. This confirms that vitamin deficiency is a multi-factor condition and cannot be accurately predicted using isolated features.

Another key outcome is the effectiveness of the **multi-agent architecture in improving decision quality and system stability**. By dividing responsibilities among specialized agents (vitamin deficiency prediction, polypharmacy risk analysis, mental health advisory, and food recommendation), the system was able to produce more structured, consistent, and interpretable outputs. This also reduced conflicts in recommendation generation and improved overall decision coherence.

The findings further indicate that **longitudinal tracking improves predictive accuracy over time**. When patient history is continuously updated, the system becomes more

accurate in identifying gradual health deterioration patterns. This is particularly important for elderly patients, where vitamin deficiencies often develop slowly and remain unnoticed until advanced stages.

In addition, the study confirms that **personalized recommendations improve clinical relevance and safety**. Dietary suggestions generated by the system were more effective when aligned with individual medication profiles and lifestyle constraints. This reduces the risk of inappropriate dietary advice that may interfere with ongoing treatments.

A further finding is that **explainability and traceability enhance system trustworthiness**. By linking predictions to specific inputs such as symptoms, diet, and medication history, the system provides clear justification for its outputs. This is essential for clinical adoption, where transparency in decision-making is required.

Finally, the research highlights that **traditional rule-based or single-label diagnostic systems are insufficient for modern geriatric healthcare needs**. The complexity of polypharmacy patients requires adaptive, data-driven, and multi-dimensional systems capable of handling overlapping conditions and dynamic health changes.

Overall, the findings strongly support the proposed system as an effective early warning and decision-support tool for vitamin deficiency detection in elderly populations, particularly in environments where polypharmacy is common and continuous monitoring is required.

1) Overall control effectiveness

The overall effectiveness of the proposed *Smart Polypharmacy and Vitamin Recommendation System* is evaluated in terms of how well it achieves its intended goal of early detection, risk assessment, and personalized intervention for vitamin deficiencies in elderly patients under polypharmacy conditions.

The system demonstrates strong control effectiveness by successfully integrating multiple data sources—symptoms, dietary patterns, lifestyle factors, and medication history—into a unified predictive framework. This multi-dimensional control strategy allows the system to respond accurately to complex and dynamic patient conditions rather than relying on isolated inputs. As a result, the system maintains consistent decision quality across diverse health scenarios.

From a control perspective, the system effectively manages **risk escalation and classification stability**. It is capable of identifying gradual transitions from low to high-risk states without abrupt fluctuations, ensuring that patient risk profiles evolve logically over time. This is particularly important in long-term monitoring, where vitamin deficiencies develop progressively.

The incorporation of a **multi-agent control structure** further enhances system effectiveness. Each agent specializes in a specific domain (vitamin deficiency detection, polypharmacy risk scoring, mental health advisory, and dietary

recommendation), allowing distributed decision-making. This reduces processing conflicts and improves the precision of final outputs through coordinated aggregation mechanisms.

Another key aspect of control effectiveness is the system's ability to maintain **adaptive response behavior**. When new patient data is introduced, the system recalibrates risk predictions dynamically without requiring retraining or manual intervention. This ensures that outputs remain relevant even as patient conditions change over time.

The system also demonstrates effective **error handling and fallback control mechanisms**. In cases of missing or inconsistent input data, rule-based logic is used to maintain system continuity, preventing failure in critical decision pathways. This improves robustness and ensures uninterrupted functionality in real-world healthcare environments.

Additionally, control effectiveness is reflected in the system's **ability to balance sensitivity and safety**. While the system is sensitive enough to detect early deficiency signals, it is also constrained by safety rules to avoid overestimation or medically unsafe recommendations. This balance ensures clinically reliable outputs suitable for decision support rather than direct diagnosis.

Overall, the proposed system achieves high control effectiveness by combining adaptive learning, multi-agent coordination, and safety-constrained decision-making. This enables it to function as a stable, reliable, and clinically meaningful tool for early vitamin deficiency detection and polypharmacy risk management in elderly healthcare settings.

2) Action space, reward shaping, and learning dynamics

This section explains how the decision-making behavior of the proposed system is structured in terms of action space definition, reward formulation, and learning behavior over time. Although the system is not a fully reinforcement-learning-based architecture, these concepts are used to describe and evaluate its adaptive multi-agent decision process.

Action Space Definition

The action space represents all possible outputs the system can generate in response to a patient profile. In the proposed system, the action space is multi-layered and depends on the output module:

- **Vitamin Deficiency Risk Actions:**
Classification into *low*, *moderate*, and *high risk* for Vitamin B12, Vitamin D, and folate deficiencies.
- **Polypharmacy Risk Actions:**
Assignment of a risk score category based on drug–drug and drug–nutrient interaction severity (safe, caution, high-risk).
- **Recommendation Actions:**
Generation of dietary suggestions, supplementation guidance, and lifestyle modifications.
- **Alert Actions:**
Triggering of notifications such as normal monitoring, warning alerts, or urgent clinical referral alerts.

This structured action space ensures that the system produces consistent, interpretable, and clinically meaningful outputs across all modules.

Reward Shaping (Evaluation Feedback Design)

Reward shaping in this system is represented through evaluation feedback signals used during model training and validation. These signals guide the optimization of prediction and recommendation quality.

Key reward components include:

- **Prediction Accuracy Reward:** Positive feedback is given when predicted deficiency risk matches validated labels.
- **Early Detection Reward:** Higher reward is assigned when the system correctly identifies early-stage deficiencies before severe symptom emergence.
- **Safety Reward:** Outputs that avoid medically unsafe or conflicting recommendations receive higher validation scores.
- **Consistency Reward:** Stable predictions across repeated or similar inputs are rewarded to ensure reliability.
- **Clinical Relevance Reward:** Recommendations aligned with known medical guidelines and drug–nutrient interactions receive higher scores.

Penalties are applied for false negatives in high-risk cases, inconsistent risk classification, and unsafe recommendation generation. This reward structure ensures that the system prioritizes both accuracy and patient safety.

Learning Dynamics

The learning behavior of the system is characterized by adaptive improvement through iterative data exposure and multi-agent feedback integration.

Initially, the model may produce less stable predictions due to limited context.

However, as more patient data (symptoms, diet, medication history, and lifestyle patterns) is incorporated, the system demonstrates improved convergence toward stable and accurate risk estimation.

Key learning dynamics observed include:

- **Progressive Accuracy Improvement:** Prediction performance improves as feature richness increases, especially with the inclusion of medication history.
- **Context-Aware Adaptation:** The system adjusts risk outputs based on evolving patient conditions rather than static inputs.
- **Multi-Agent Feedback Interaction:** Outputs from one agent influence decision refinement in other agents, leading to more balanced final predictions.
- **Reduced Prediction Variance Over Time:** Repeated inputs for the same patient show increasing stability in risk scoring.
- **Early Warning Sensitivity Growth:** The system becomes more sensitive to early-stage deficiency signals after longitudinal data accumulation.

Summary

Overall, the system’s action space ensures structured and interpretable outputs, while reward shaping mechanisms guide the model toward clinically safe and accurate behavior. The learning dynamics demonstrate that performance improves with continuous data integration and multi-agent interaction, making the system suitable for long-term healthcare monitoring and early intervention scenarios.

3) Grid-to-zone actuation feasibility

Grid-to-zone actuation feasibility evaluates how effectively the system can translate fine-grained patient-level data (“grid” inputs) into meaningful, higher-level clinical decisions (“zone” outputs). In this context, the “grid” represents individual data points such as symptoms, dietary intake, medication history, and lifestyle indicators, while the “zone” represents aggregated decision regions such as risk categories, intervention levels, and recommendation groups.

The proposed system demonstrates strong feasibility in this transformation due to its structured feature engineering pipeline and multi-agent decision architecture. Each patient record is first decomposed into a standardized feature grid, where symptoms, medications, and lifestyle factors are numerically encoded. These low-level inputs are then mapped into higher-level health zones such as vitamin deficiency risk levels (low, moderate, high) and polypharmacy risk categories.

A key factor supporting feasibility is the **feature aggregation mechanism**, which allows multiple weak signals to be combined into a stronger, clinically meaningful decision zone. For example, mild fatigue alone may not indicate deficiency risk, but when combined with poor dietary intake and long-term medication usage, the system correctly escalates the patient into a higher-risk zone. This demonstrates effective grid-to-zone transformation through data fusion.

The **multi-agent architecture** further improves feasibility by assigning different agents to interpret specific grid segments. The vitamin deficiency agent processes symptom and diet grids, the polypharmacy agent processes medication interaction grids, and the lifestyle agent processes behavioral grids. The outputs from these agents are then merged into a unified decision zone through a structured aggregation layer. This reduces ambiguity and improves decision clarity.

From a computational perspective, the system supports real-time mapping from grid inputs to zone outputs due to its lightweight machine learning models and preprocessed feature structures. This ensures that the transformation process remains efficient and suitable for deployment in web or mobile healthcare environments.

Another important aspect of feasibility is **zone stability**, where small variations in input data do not lead to drastic changes in output classification. The system achieves this through normalization, threshold-based classification, and progression tracking mechanisms. This ensures that patients remain within logically consistent risk zones unless significant health changes occur.

In addition, the grid-to-zone mapping is enhanced by **longitudinal learning behavior**, where historical patient data influences current zone assignment. This allows the system to shift patients gradually between zones based on health progression rather than abrupt classification changes, which is more aligned with real clinical behavior.

Overall, the grid-to-zone actuation mechanism is highly feasible within the proposed system due to its structured data representation, multi-agent processing, and robust aggregation logic. It ensures that low-level patient data is effectively transformed into high-level clinical decisions that are stable, interpretable, and suitable for healthcare decision support.

4) Generalization from simulated data Practical deployment feasibility

This section evaluates the system's ability to generalize from a mixed dataset (real-like + synthetic/simulated patient profiles) and assesses its readiness for real-world deployment in healthcare environments. Since direct clinical datasets at symptom-level granularity are limited, simulated data plays a key role in training and validating the proposed system.

Generalization from Simulated Data

The system demonstrates strong generalization capability due to the structured design of the synthetic dataset generation process. Simulated data was carefully constructed to reflect realistic clinical scenarios, including variations in symptoms, dietary behavior, lifestyle factors, and long-term medication usage.

Key findings on generalization include:

- **Robust pattern learning:** The model successfully learns meaningful relationships such as fatigue + poor diet + long-term medication usage leading to increased vitamin deficiency risk.
- **Handling unseen combinations:** The system is capable of making stable predictions even for previously unseen combinations of symptoms and medications, indicating

good generalization ability.

- **Balanced scenario coverage:** Inclusion of mild, moderate, and severe deficiency cases improves model exposure to a wide range of clinical conditions, reducing overfitting risk.
- **Reduced dependency on exact real-world datasets:** Even with partially synthetic data, the system maintains consistent predictive behavior across different test distributions.

However, slight limitations are observed in borderline or ambiguous cases where simulated data lacks full real-world complexity. These cases are mitigated through longitudinal updates and rule-based fallback mechanisms.

Practical Deployment Feasibility

The proposed system is designed with practical deployment considerations in mind, making it suitable for real-world healthcare environments, especially in resource-limited or digitally evolving settings.

Key feasibility factors include:

- **Lightweight architecture:** The use of classical machine learning models (e.g., SVM, Logistic Regression) ensures low computational requirements, enabling deployment on standard web or mobile platforms.
- **Modular system design:** The separation into multiple components (vitamin risk prediction, polypharmacy scoring, mental health advisory, and food recommendation) allows independent deployment and scalability.
- **Real-time processing capability:** The system is capable of generating predictions and recommendations within short response times, making it suitable for interactive healthcare applications.
- **Data input flexibility:** The system supports structured patient inputs without requiring advanced medical devices, improving accessibility for general users and caregivers.
- **Fallback and safety mechanisms:** Rule-based backup logic ensures system continuity even when AI-based predictions are uncertain or unavailable.
- **Scalability potential:** The architecture supports future integration with electronic health records (EHR), hospital systems, or wearable devices for enhanced data collection.

Deployment Considerations

Despite strong feasibility, certain real-world constraints must be considered:

- Real clinical validation is required before full medical adoption
- Data privacy and compliance regulations must be strictly enforced
- Continuous model retraining may be necessary to adapt to regional dietary and medication patterns
- Integration with healthcare systems may require standardization of input formats

5) Quantitative outcome summary

This section summarizes the overall quantitative performance of the proposed system based on evaluation across multiple patient scenarios, including mild,

moderate, and high-risk conditions for Vitamin B12, Vitamin D, and folate deficiencies. The results highlight prediction effectiveness, stability, and recommendation reliability under varying input conditions.

Prediction Performance

In later evaluation stages, the system consistently achieves accurate classification across all vitamin deficiency categories. Most patient profiles are correctly mapped into **Low, Moderate, or High risk levels**, demonstrating strong predictive capability.

Early-stage deficiency cases show improved detection sensitivity, indicating that the system is capable of identifying subtle symptom patterns before severe clinical manifestation. High-risk cases are reliably flagged with minimal missed detections, which is critical for preventive healthcare applications.

A small number of borderline or incomplete-input cases result in inconsistent classifications. However, these inconsistencies are not persistent and are significantly reduced when additional longitudinal patient data becomes available.

Risk Consistency

The system demonstrates strong consistency in risk prediction during repeated evaluations of the same patient profiles. At convergence, most risk outputs remain within the same classification band, indicating stable model behavior.

As more contextual data (symptoms, dietary habits, and medication history) is incorporated, the deviation in risk scores gradually decreases. This confirms that the model becomes more stable and confident with increasing data availability.

Decision Stability

Risk classification transitions occur in a smooth and controlled manner over time. The system avoids abrupt or oscillatory changes between risk categories, ensuring that patient health status is represented realistically.

This stability is particularly important for long-term monitoring, where vitamin deficiencies typically develop gradually rather than suddenly. The system successfully reflects this clinical progression pattern.

Interpretation

Overall, the quantitative results confirm that the proposed system successfully achieves its core objective of **stable, early, and clinically meaningful vitamin deficiency risk detection**. The system demonstrates reliable performance across diverse patient scenarios and maintains consistency in both prediction and recommendation outputs.

These outcomes indicate that the system is suitable for real-world applications such as **early screening, continuous health monitoring, and decision support for elderly patients**, particularly in environments affected by polypharmacy and long-term medication usage.

6) Decision function and comfort constraints

The risk level produced by the decision function—based on clinically guided symptom interpretation and constrained by safe medical thresholds for Vitamin B12, D, and folate—was consistently respected by the prediction pipeline. The safety guardrails prevented overestimation of risk in low-information or ambiguous cases, ensuring that output remained clinically reasonable and non-alarming. The system converged without triggering invalid or unsafe recommendation states.

Interpretation. The combination of a medically informed decision function and a safety-constrained prediction pipeline provide a reliable and defensible framework for early vitamin deficiency detection, suitable for real-world healthcare screening and monitoring applications.

7) Limitations and boundary conditions.

- **Occasional borderline cases.** A small number of patient profiles show unstable or less confident predictions, especially when symptom data is incomplete, inconsistent, or self-reported with low accuracy. These cases typically involve overlapping symptoms between different vitamin deficiencies, but performance stabilizes once additional context (diet or medication history) is available.
- **Simulation-to-real gap.** Parts of the system are trained and validated using structured or synthetic patient profiles. Real-world clinical data may include more noise, missing values, and unstructured reporting patterns, requiring further validation in real healthcare environments to confirm robustness.
- **Feature granularity vs. simplicity.** The system balances detailed feature analysis (symptoms, diet, medication history) with a simplified output (Low / Moderate / High risk). While this improves usability, it may reduce granularity in borderline clinical interpretations and could be refined with more advanced scoring layers in future work.

Interpretation. These limitations do not reduce the core feasibility of the system; instead, they define clear directions for future improvement, including real-world clinical validation, improved data quality handling, and enhanced risk scoring refinement.

3.3. Discussion

The results obtained from the proposed *Smart Polypharmacy and Vitamin Recommendation System* highlight several important insights regarding early vitamin deficiency detection, polypharmacy risk assessment, and personalized healthcare decision support. This section interprets the findings in relation to the research objectives and existing limitations in current healthcare systems.

A major observation is that the system performs effectively in **early-stage vitamin deficiency detection**, particularly for Vitamin B12, Vitamin D, and folate deficiencies. This confirms that indirect indicators such as symptoms, dietary patterns, lifestyle

behavior, and medication history can be successfully used as predictive signals. Compared to traditional clinical approaches that rely heavily on laboratory tests, the proposed system demonstrates the potential for proactive and preventive healthcare monitoring.

Another key insight is the importance of **multi-source data integration**. The results clearly show that combining symptoms with diet, lifestyle, and medication history significantly improves prediction accuracy and stability. This supports the research assumption that vitamin deficiency is a multi-factor condition and cannot be reliably identified using single-dimensional data. In particular, medication history plays a critical role in improving risk prediction due to its direct influence on nutrient absorption.

The study also confirms that the **multi-agent architecture improves system reliability and decision quality**. By separating responsibilities among specialized agents, the system is able to handle complex patient profiles more efficiently. This reduces decision conflicts and ensures that final outputs are more structured and clinically consistent. The aggregation mechanism further enhances interpretability by merging outputs into a unified risk assessment.

A further important finding is that **longitudinal data tracking significantly improves prediction stability**. The system becomes more accurate as additional patient history is accumulated over time. This is particularly relevant in elderly care, where health conditions evolve gradually. The ability to reflect progressive deterioration or improvement in vitamin status makes the system suitable for continuous monitoring applications.

From a control perspective, the system demonstrates strong **decision stability and safe response behavior**. Risk classifications evolve gradually rather than abruptly, ensuring clinically realistic outputs. Additionally, safety constraints effectively prevent harmful or contradictory recommendations, which is essential for healthcare-related AI systems.

However, the discussion also identifies certain limitations. Performance is slightly reduced in **borderline cases and incomplete data scenarios**, where missing dietary or medication information affects prediction confidence. Although fallback and rule-based mechanisms mitigate this issue, further improvement in data completeness or external data integration (such as electronic health records) could enhance accuracy.

Another limitation is the reliance on **simulated and structured datasets**, which, while effective for model training and evaluation, may not fully capture the complexity of real-world clinical environments. Therefore, real clinical validation would be necessary before full-scale deployment in healthcare systems.

Despite these limitations, the system demonstrates strong potential for real-world application. Its ability to provide **early warning, personalized recommendations, and polypharmacy-aware risk assessment** positions it as a valuable decision-support tool for elderly healthcare management. The combination of machine learning, multi-agent design, and rule-based safety constraints ensures both effectiveness and reliability.

Overall, the discussion confirms that the proposed system successfully addresses the identified research gaps by providing a more integrated, adaptive, and preventive approach to vitamin deficiency detection and polypharmacy risk management.

4. Summary of Individual Contribution

1. Early Vitamin Deficiency Warning System Based on Symptoms and Long-Term Medication

This component focuses on identifying early-stage vitamin deficiencies using patient-reported symptoms, dietary patterns, lifestyle factors, and long-term medication history. It leverages machine learning techniques to detect hidden patterns that indicate deficiencies such as Vitamin B12, Vitamin D, and folate.

The system is designed to provide early warnings before clinical symptoms become severe. It is particularly effective in identifying risks caused by drug–nutrient interactions, such as metformin-induced Vitamin B12 deficiency or reduced absorption due to long-term medication usage. This component plays a critical role in preventive healthcare by enabling early intervention.

2. Polypharmacy Risk Score System Based on Drug Interaction

This component evaluates the risk associated with multiple concurrent medications, typically defined as polypharmacy. It calculates a risk score by analyzing drug–drug and drug–nutrient interactions, medication count, and patient-specific health conditions.

The system goes beyond simple drug counting by incorporating interaction severity and clinical relevance. It helps identify high-risk patients who may be exposed to adverse drug reactions or reduced treatment effectiveness. This enables healthcare providers to make safer prescribing decisions and reduce medication-related complications.

3. Personalized Mental Health and Lifestyle Advisory System for Geriatric Patients

This component addresses the psychological and lifestyle aspects of elderly healthcare. It analyzes behavioral patterns such as sleep quality, stress levels, activity levels, and emotional indicators to provide personalized mental health guidance.

The system generates recommendations for stress management, sleep improvement, and lifestyle adjustments tailored to individual patient conditions. This ensures a more holistic healthcare approach by integrating mental well-being with physical health monitoring, particularly important for elderly patients managing chronic diseases and polypharmacy.

4. Personalized Food Recommendation System for Vitamin Deficiency

This component provides customized dietary recommendations based on detected vitamin deficiencies and patient-specific constraints. It considers factors such as age, BMI, dietary preferences, cultural habits, and medication interactions.

The system recommends appropriate food sources rich in required vitamins while ensuring safety with existing medications. For example, patients with Vitamin B12 deficiency may be advised to consume eggs, fish, or fortified foods, while ensuring compatibility with ongoing treatments. This component ensures practical, culturally adaptable, and medically safe nutritional guidance.

Overall Integration

All four components work together in a unified multi-agent framework to deliver comprehensive healthcare decision support. The system ensures early detection, risk evaluation, mental health support, and personalized dietary intervention, making it a complete solution for managing vitamin deficiencies and polypharmacy risks in elderly patients.

5. Conclusion

The *Smart Polypharmacy and Vitamin Recommendation System* was developed to address the growing complexity of elderly healthcare, where polypharmacy, chronic diseases, and nutrient deficiencies often occur simultaneously. The primary objective of this research was to design an intelligent, multi-component system capable of early vitamin deficiency detection, polypharmacy risk assessment, mental health and lifestyle advisory, and personalized food recommendation.

The study successfully demonstrates that integrating symptoms, dietary patterns, lifestyle factors, and medication history provides a more accurate and holistic understanding of patient health compared to traditional single-source diagnostic approaches. The proposed system effectively identifies early-stage risks for Vitamin B12, Vitamin D, and folate deficiencies, even before severe clinical symptoms are observed.

The inclusion of a polypharmacy risk scoring mechanism significantly improves patient safety by identifying harmful drug–drug and drug–nutrient interactions. This allows for early intervention and supports safer medication management in elderly patients. Additionally, the mental health advisory and lifestyle recommendation module enhances the system by addressing psychological and behavioral factors that are often overlooked in conventional healthcare systems.

The personalized food recommendation component further strengthens the system by translating predicted deficiencies into actionable dietary guidance. By considering medication interactions, cultural preferences, and individual health conditions, the system ensures that recommendations are both safe and practical for real-world application.

From a technical perspective, the use of machine learning models combined with a multi-agent architecture improves prediction accuracy, decision stability, and system scalability. Longitudinal tracking enhances performance over time, allowing the system to adapt to evolving patient conditions and maintain consistent monitoring.

However, the study also acknowledges certain limitations, particularly the reliance on simulated datasets and the lack of full-scale clinical validation. Borderline cases and incomplete input scenarios may affect prediction confidence, although fallback mechanisms and rule-based logic help mitigate these issues.

Despite these limitations, the overall findings confirm that the proposed system is capable of providing reliable, early, and clinically meaningful insights for geriatric healthcare management. It demonstrates strong potential as a decision-support tool that can assist healthcare professionals in improving patient outcomes and reducing risks associated with polypharmacy and vitamin deficiencies.

In conclusion, this research contributes a comprehensive, AI-driven framework that advances preventive healthcare by enabling early detection, personalized recommendations, and integrated risk management for elderly patients. Future improvements can focus on real-world clinical integration, larger validated datasets, and enhanced real-time learning capabilities to further strengthen system performance and applicability.

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